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Measurements of Fine Particle Mass Concentrations Using Continuous and Integrated Monitors in Eastern US Cities

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$PM_{2.5}$ mass concentrations were measured using multiple continuous samplers (a continuous ambient mass monitor (CAMM), a real-time ambient mass sampler (RAMS), and a drier-equipped tapered element oscillating microbalance (TEOM)) as well as a federal reference method (FRM) filter sampler in three eastern US cities. The effects of measurement systems, ambient conditions, and composition on the measured $PM_{2.5}$ mass concentrations were examined at each site. Different responses to $PM_{2.5}$ mass were observed by the continuous samplers with a variation of 10–30% in the average PM mass concentrations. The impacts of semivolatile particulate ammonium nitrate on $PM_{2.5}$ mass concentration was apparent for those time periods when high nitrate concentration >1.62 g m^{-3} was measured in Atlanta. Contribution of semivolatile particulate ammonium nitrate to the ambient $PM_{2.5}$ mass appeared more significant than semivolatile organic material in Atlanta. The effects of the ambient water were not negligible and were significant in Baltimore. The total ambient $PM_{2.5}$ mass by the CAMM was underestimated in all locations compared to the other continuous PM mass monitors, presumably due to the lack of equipments that collect semivolatile particulate ammonium nitrate and organic materials. The results suggest that short time intervals do not guarantee the measurement of total ambient $PM_{2.5}$ mass including semivolatile materials. In the comparison of each of the continuous versus integrated $PM_{2.5}$ mass over the sites, the drier-equipped TEOM presented the best agreement, with the FRM having r^2 values >0.90 and slopes close to 1 regardless of location and time period, suggesting that the same losses of semivolatile materials as seen in the FRM can occur in the sampling by the drier-equipped TEOM. The regressions between the CAMM or the RAMS and the FRM showed the variance of their relation, by site and time period, from 0.6 to 0.9 for r^2 and from 0.7 to 1.0 for the slope.

INTRODUCTION

The US Environmental Protection Agency (USEPA) promulgated National Ambient Air Quality Standards (NAAQS) in 1997 for fine particulate matter with a diameter less than 2.5 μm ($PM_{2.5}$) in order to protect public health from air pollutants and mandated the states monitor to determine compliance with these standards. The federal reference method (FRM) was designed by the USEPA to collect $PM_{2.5}$ mass in order to determine the attainment and nonattainment area of interest for the NAAQS. The FRM is an integrated filter-based measurement method collecting particulate matter (PM) on a 24 h basis. However, the atmospheric aerosol is composed of nonvolatile (NV) and semivolatile (SV) inorganic and organic materials that can be exchanged between the particulate and gaseous phases. Thus, during long sampling and postsample intervals, volatilization of semivolatile compounds from particles and/or absorption of organic gases onto particles may occur, possibly leading to inaccurate observation of airborne $PM_{2.5}$ mass concentration (Hering and Cass 1999). The losses due to volatilization may be prominent in urban areas where semivolatile inorganic and organic compounds are abundant in fine particles. It is important to measure both the NV and SV components of PM in order to determine more accurately the relationship between PM and various health and welfare effects.

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Two real-time samplers were recently developed with the intention of measuring both the NV and SV components of $PM_{2.5}$ concentrations: the continuous ambient mass monitor (CAMM; Babich et al. 2000a) and the real-time total ambient mass sampler (RAMS; Eatough et al. 1999b, 2001; Pang et al. 2002). The CAMM uses the pressure difference across a membrane filter that is proportional to $PM_{2.5}$ concentration. In the RAMS, diffusion denuder technology and a particle concentrator are combined with dual tapered element oscillating microbalances (TEOMs). Both the CAMM and RAMS can collect fine particles for time intervals less than 1 h.

In order to examine the utility of these samplers, a series of intensive $PM_{2.5}$ measurements have been conducted in three eastern US cities, Philadelphia, Baltimore, and Atlanta, using the CAMM and RAMS, along with a TEOM with a Nafion drier and nephelometers. Because there is no method to determine the true PM mass, the $PM_{2.5}$ mass concentrations for the various methods were compared with the other methods to evaluate the performance of each method.

Comparisons with the FRM $PM_{2.5}$ mass concentrations were also performed to determine the amounts and proportions of semivolatile compounds that are not measured by the FRM within the true airborne $PM_{2.5}$ concentration.

MEASUREMENTS

Site Description

The Atlanta site (33.7767 N and 84.4166 W, elevation 275 m) on Jefferson Street was located in commercial/industrial/residential area and has been operated since 1998 as part of the Southeastern Aerosol Research and Characterization (SEARCH) air-monitoring network. The data were collected during November and December 2001. The site, 4.2 km northwest of downtown Atlanta, GA, was on a grassy knoll surrounded by parking lots, city streets, warehousing, and storage buildings (Hansen et al. 2003).

The Philadelphia site (40.0372 N, 75.0078 W) was at the Baxter Water Treatment Center in northern Philadelphia, PA. This has been the site for the North American Research Strategy for Tropospheric Ozone (NARSTO)—Northeast Oxidant and Particle Study (NEOPS). The site was in a grass-covered field surrounded on three sides by thin lines of mixed deciduous and pine trees, with the other side being bounded by the Delaware River. The immediate surrounding area consisted of corporate and industrial facilities and the northeastern suburbs of Philadelphia. The site was approximately 0.5 km southeast of Interstate 95 and 2 km southwest of the Bucks County, PA county line. Downtown Philadelphia and downtown Trenton, NJ are within a 30 km radius of the site. During the study period, hot summer conditions prevailed with daytime high temperatures above 30°C and high relative humidity (RH) of 80% on average.

The Baltimore supersite trailer was installed at the FMC Corporation site (39.2303 N, 76.5803 W) near Brooklyn, MD. The FMC Corp. is a chemical manufacturing facility located at 1701 E. Patapsco Avenue in a heavily industrialized area of Baltimore. This area has been the location of chemical industrial activity since 1915. To the north, northwest, and east of the site are other manufacturing companies. Curtis Bay and Stonehouse Cove define the southern and southwestern boundaries, respectively. Baltimore is a populous, industrial deepwater port city, located 50 km north of Washington, DC and 150 km east of the Appalachian mountains. As a major traffic thoroughfare, the air quality of Baltimore is heavily influenced by transported aerosol emissions from the Ohio River valley and Washington, DC, as well as by local industrial and heavy diesel vehicle emissions.

Methods

A series of $PM_{2.5}$ measurements were conducted at Atlanta, GA from 21 November to 23 December 2001; at Philadelphia, PA from 2 July to 1 August 2001; and at Baltimore, MD from 17 May to 11 June 2001. Hourly $PM_{2.5}$ mass concentrations were measured continuously using the CAMM, the RAMS, and the TEOM with a Nafion drier at each site.

The CAMM, developed at the Harvard School of Public Health, collects ambient particles on a short time (typically 30–60 min) basis to minimize volatilization and absorption artifacts due to changes in concentration and composition during sampling (Babich et al. 2000a) and now is a commercial unit supplied by Thermo Andersen, Smyrna, GA. Subsequently it was removed from the market. The measurements of the system are based on the determination of the increase in pressure drop across a Teflon membrane filter (Fluoropore^{TM}) during particle sampling. For each sampling period, a new segment of the filter tape was exposed to the airstream so that particles would remain close to equilibrium with the sample air during their collection. The CAMM employs a Nafion drier to remove particle-bound water. A detailed description can be found in Babich et al. (2000a).

The RAMS is a system consisting of a particle concentrator—Brigham Young University organic sampling system (PC-BOSS) combined with dual TEOM technology. The particle concentrator (PC) in the RAMS is a slotted virtual impactor with a 50% particle cutpoint at about 0.1 μm (Sioutas et al. 1994). A concentrated stream of 0.1–2.5 μm is provided based on the PC with the minor flow stream (typically 10–25% of the total flow), and the major flow stream is discarded (Eatough et al. 1999b). The PC efficiency was calculated by the ratio of the sulfate mass in the minor flow stream to the sum of the sulfate mass in the minor and major flow streams (Eatough et al. 1999b). The PC efficiency (E_{pc}) of the RAMS was 0.82 for Atlanta and 0.68 for Philadelphia. The efficiency of the RAMS in Atlanta appeared to be higher than the suggested values

(0.5–0.7) by Eatough and coworkers (Eatough et al. 1999a; Ding et al. 2002). Therefore, the adjustment was made for the Atlanta RAMS values using a typical concentrator efficiency suggested. Here, a value of 0.5 was used as the typical value, and the adjusted RAMS mass concentrations were used throughout this article. The RAMS mass concentration was adjusted by multiplying a factor of 1.64 (the ratio of the observed E_{pc}, 0.82, to the typical E_{pc}, 0.5) to the original RAMS mass concentrations. The same efficiency value was also used for the Atlanta supersite studies (Modey et al. 2001). No particle concentrator was used in Baltimore to explore the operation of the system without concentration of the ambient PM. Detailed description for the system is presented by Eatough and coworkers (Eatough et al. 1999b, 2001; Ding et al. 2002; Pang et al. 2002).

The TEOM used in this study was the TEOM with a sample equilibrium system (i.e., SES TEOM) and modified with a Nafion dryer (Perma Pure, Toms River, NJ) attached in the inlet air stream (Eatough et al. 2001), in order to condition the PM sample stream to a lower humidity and temperature level (Meyer et al. 2000). The SES TEOM (Model 1400 a/b; Patashnick and Rupprecht 1991) in Atlanta, modified by atmospheric research and analysis (ARA), was operated at 30°C. In Philadelphia and Baltimore, the operating temperature of the SES TEOM was set to 35°C because it was too unstable at 30°C and sometimes the ambient temperature was higher than 30°C. Two nephelometers (Radiance Research M903) with and without a Nafion dryer (Perma Pure) were also used in the measurements.

A variety of continuous and semicontinuous samplers for $PM_{2.5}$ speciation were deployed at each site. In Atlanta, the ARA denuder difference technique was used to measure 1 h average particulate nitrate, including semivolatile material (SVM). Ambient air passes through a series of denuders to remove gaseous HNO_3, oxides of nitrogen, and NH_3, and particulate nitrate was detected via O_3 chemiluminescence with conversion to NO on a 350°C molybdenum (Mo) converter. A detailed description of this nitrate measurement is provided by (Weber et al. 2003). Organic carbon (OC) concentrations were measured using a R&P 5400 ambient carbon particulate monitor with impactor-style collectors, operated with two temperature plateaus (275°C for 480 s and 750°C for 240 s). After passing through the inlet and sample line, particles are collected on a first impactor plate for approximately 1 h. Then the flow path is automatically switched to collect particles on a second impaction plate. Meanwhile, the first plate is analyzed, and the second plate is analyzed while the first plate returns to sampling.

In Philadelphia, particulate nitrate was measured every 15 min with a system described by Simon and Dasgupta (1993, 1995). One-hour average total carbon (TC) and optical elemental carbon (EC) concentrations were measured using Sunset thermal EC/OC analyzer (Sunset Laboratory, Tigard, OR, USA), and OC concentrations were calculated by subtracting optical EC concentrations from TC concentrations.

In Baltimore, continuous nitrate and OC concentrations were measured using the R&P 8400N (Stolzenburg and Hering 2000) every 10 min and using the R&P 5400 every 1 h, respectively. Continuous measurement values were averaged to 1 h intervals and also to 24 h average values to compare with 24 h integrated $PM_{2.5}$ concentrations.

It should be noted that there are potential problems with the continuous carbon monitors. The samplers used to measure nitrate were developed to measure particulate ammonium nitrate, including the SVM. However, the Sunset carbon monitor collects particles on a quartz filter before sample analysis and is vulnerable to the same semivolatile organic material (SVOM) losses as any single quartz filter sampler. It also uses a denuder prior to sample collection to remove gas-phase SVOM. Hence, it will not be expected to collect SVM. The R&P carbon monitor uses a heated impactor plate and is also expected not to collect SVM.

Twenty-four-hour average integrated concentrations of $PM_{2.5}$ mass and species were also measured at these sites. The R&P Partisol-Plus sequential air sampler (Model 2025) was operated during the study period at the Atlanta site for the $PM_{2.5}$ FRM measurement. Integrated samples of $PM_{2.5}$ species in Atlanta were collected using the three-channel filter-based particle composition monitor (PCM; Kim et al. 2003). In Philadelphia, a Harvard impactor (HI) was used to collect $PM_{2.5}$ mass concentration as an equivalent to FRM measurements (Babich et al. 2000b). A Harvard-EPA Annular Denuder System (HEADS) was also used to measure $PM_{2.5}$ species in the daytime and nighttime for 10 h intervals in Philadelphia. The description of the HEADS can be found elsewhere (Koutrakis et al. 1988). Twenty-four-hour average $PM_{2.5}$ mass concentrations (i.e., FRM) in Baltimore were measured using a reference ambient air sampler (RAAS-100; Thermo Electron Corp.) $PM_{2.5}$ system. No 24 h average $PM_{2.5}$ speciation samples were measured at the Baltimore site.

For the light-scattering coefficients, two standard integrating portable nephelometers (Radiance Research Model M903) were used with and without a Nafion drier. They measured the light-scattering extinction coefficient using the defining optical filter (530 nm). The nephelometers were run inside the trailer at the sites, but the sampling inlet was exposed to the ambient air. Ambient air was drawn into the nephelometer through the manifold. The Nafion drier was placed between the manifold and the nephelometer to avoid water condensation on particles. No size-segregating inlet to provide a 2.5 μm size-cut was used in the nephelometers since almost all light scattering is presumably due to particles smaller than 2.5 μm (Butcher and Charlson 1972).

RESULTS AND DISCUSSION

Continuous $PM_{2.5}$ Measurements

Continuous $PM_{2.5}$ mass concentrations measured at the Atlanta site are shown in Figure 1, along with semivolatile

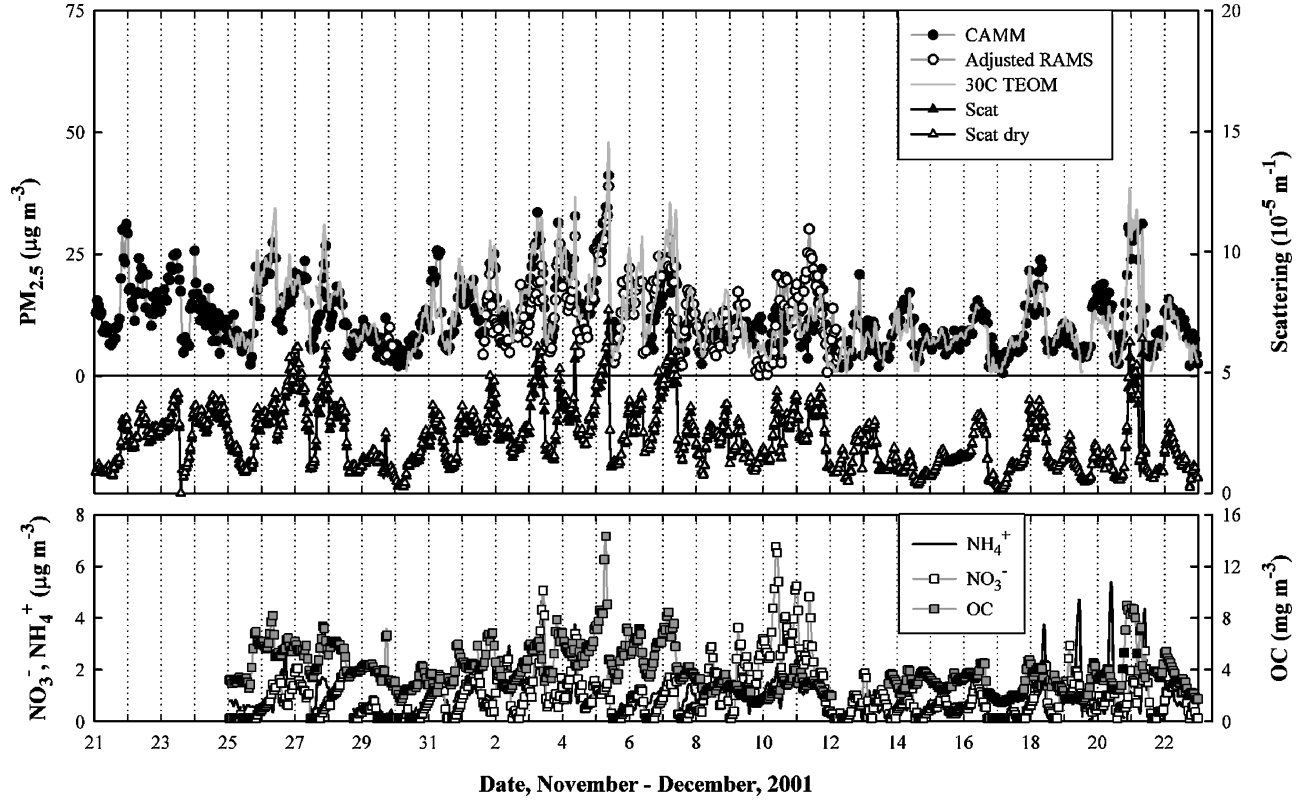


FIG. 1. One hour measurements of concentrations of $PM_{2.5}$ mass and semivolatile nitrate and OC in Atlanta, GA, along with particle light scattering. The RAMS mass concentrations were adjusted by a factor of 1.64.

$PM_{2.5}$ species variation. Table 1 presents mean, maximum, and minimum concentrations for valid $PM_{2.5}$ mass samples of the continuous samplers at each site. After quality control, the average concentration ± standard error of valid hourly $PM_{2.5}$ mass in Atlanta was 12.0 ± 0.24 μg m^{-3} (N = 760) for the CAMM, 17.9 ± 0.63 μg m^{-3} (N = 242) for the adjusted RAMS, and 12.4 ± 0.30 μg m^{-3} (N = 645) for the SES TEOM. During the study period, a relatively weak covariance was observed for $PM_{2.5}$ mass concentrations between the CAMM and adjusted RAMS (r^2 = 0.45) and between the SES TEOM and adjusted RAMS (r^2 = 0.54), whereas a strong convariance (r^2 = 0.79) was shown between the SES TEOM and CAMM $PM_{2.5}$ mass. A total of 196 contemporaneous samples from each method were available for intersampler comparison of the concentrations.

TABLE 1
Summary of $PM_{2.5}$ mass concentrations (μg m^{-3}) of the continuous samplers at the sites

	CAMM				RAMS[a]				SES TEOM[b]			
	N	Min	Mean	Max	N	Min	Mean	Max	N	Min	Mean	Max
Atlanta, GA (11/21–12/23, 2001)	760	0.60	12.0	41.2	242	0.26	17.9	52.8	645	1.00	12.4	47.5
Philadelphia, PA (7/2–8/1, 2001)	712	0.13	10.3	44.5	200	2.20	10.6	45.2	533	1.02	14.2	48.6
Baltimore, MD (5/17–6/11, 2001)	331	0.85	15.1	56.1	279	1.30	21.8	67.9	331	1.45	18.7	74.4

[a]The RAMS in Atlanta was adjusted by a factor of 1.64.
[b]The operating temperature of the SES TEOM was set to 30°C in Atlanta, but to 35°C in Philadelphia and Baltimore.

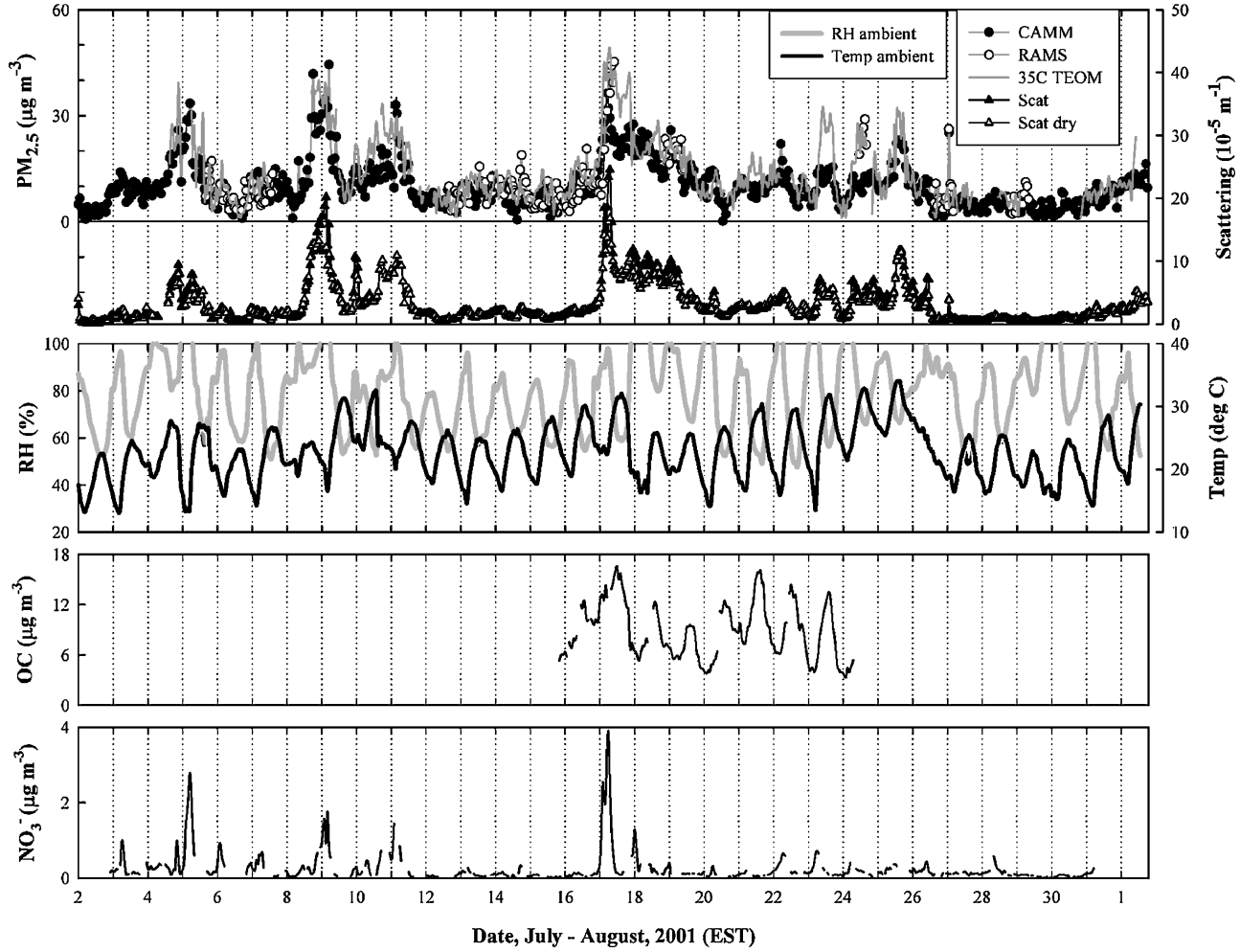


FIG. 2. One hour measurements of concentrations of $PM_{2.5}$ mass and semivolatile nitrate and organic carbon in Philadelphia, PA, along with particle light scattering and meteorological parameters.

For these 196 samples, the mean concentration ± standard error of the mean was observed to be 14.2 ± 0.5 μg m^{-3} for the CAMM, 18.6 ± 0.7 μg m^{-3} for the adjusted RAMS, and 15.5 ± 0.6 μg m^{-3} for the SES TEOM. The SES TEOM mass showed good agreement with the light scattering measured with the nephelometers ($r^2 = 0.80$), whereas both the CAMM and adjusted RAMS showed a relatively mild agreement ($r^2 = 0.64$ and 0.63, respectively).

Figure 2 shows the Philadelphia results, which presented a generally good covariance with r^2 ranging from 0.55 to 0.70 between the continuous $PM_{2.5}$ mass samplers, although some discrepancy was observed for the SES TEOM. The mean concentration of the valid hourly CAMM $PM_{2.5}$ samples was 10.3 ± 0.25 μg m^{-3} (N = 712) and that for the RAMS was 10.6 ± 0.54 μg m^{-3} (N = 200), which were lower than that for the SES TEOM, 14.2 ± 0.40 μg m^{-3} (N = 533). Among these samples, 185 contemporaneous samples were available for intersampler comparison. The mean $PM_{2.5}$ mass concentrations of each sampler for these 185 samples were 9.1 ± 0.46 μg m^{-3} for the CAMM, 10.9 ± 0.58 μg m^{-3} for the RAMS, and 11.8 ± 0.65 μg m^{-3} for the SES TEOM. Scattering coefficients without drier covaried very well with scattering coefficients with drier ($r^2 = 0.95$) and tracked well with $PM_{2.5}$ mass concentrations ($r^2 = 0.78$ with the CAMM, 0.63 with the RAMS, and 0.72 with the SES TEOM).

$PM_{2.5}$ mass concentrations (Figure 3) at the Baltimore site showed the least covariance between the continuous methods, which is quite different from the other two sites. In addition, the measured light scattering did not follow the variation of $PM_{2.5}$ mass concentrations well. However, the correlation of scattering coefficients between the dried and nondried nephelometers appeared good ($r^2 = 0.75$). The mean concentration of valid hourly $PM_{2.5}$ mass concentration samples was 15.1 ± 0.55 μg m^{-3} (N = 331) for the CAMM, 21.8 ± 0.74 μg m^{-3} (N = 279) for the RAMS, and 18.7 ± 0.63 μg m^{-3} (N = 331) for the SES TEOM. For the 120 samples that were available from

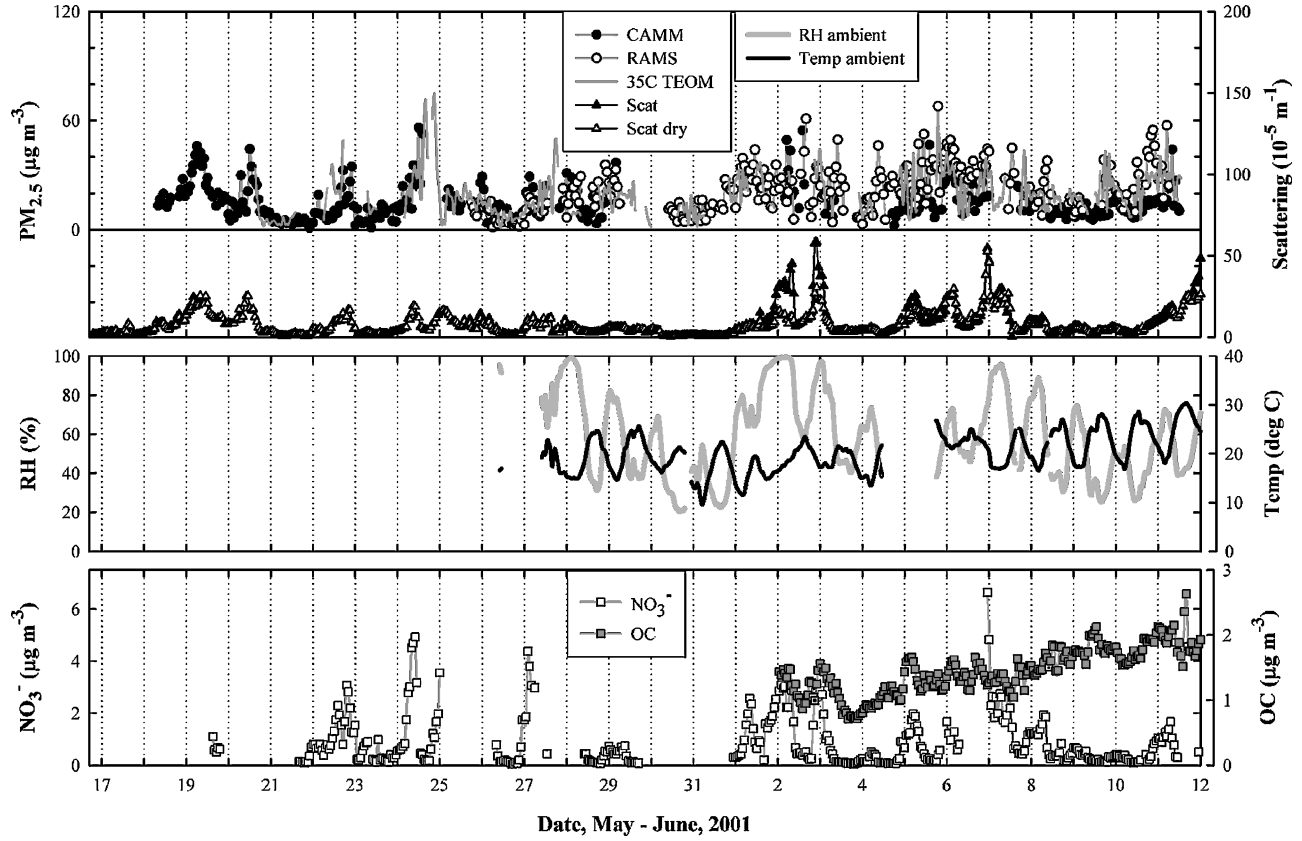


FIG. 3. One hour measurements of concentrations of $PM_{2.5}$ mass and semivolatile nitrate in Baltimore, MD, along with particle light scattering and meteorological parameters.

each monitor, the mean $PM_{2.5}$ mass concentration was 15.8 ± 0.8 μg m^{-3} for the CAMM, 22.5 ± 1.2 μg m^{-3} for the RAMS, and 19.5 ± 0.9 μg m^{-3} for the SES TEOM.

The maxima and minima of $PM_{2.5}$ mass of the continuous monitors given in Table 1, along with 12 h moving averaged PM mass shown in Figure 4, describes the dynamic ranges of $PM_{2.5}$ mass concentrations at each site. Here, the dynamic range defines the range between the maximum and minimum values and the rate of change. Since the measurements were conducted during different periods, a direct comparison across the three sites is not feasible. Philadelphia and Baltimore might have similar dynamic ranges because samples at both sites were taken during summer of 2001. Table 1 shows that the range of peak of $PM_{2.5}$ mass in Philadelphia was similar among the continuous methods compared to the ranges at the other sites. A wider range was observed in Baltimore among the continuous methods. This wide range (0.85–74.4) was not seen in the other two sites (i.e., Atlanta and Baltimore). Similar ranges were observed in Atlanta (0.26–52.8) and Philadelphia (0.13–48.6).

Intersampler Comparison of $PM_{2.5}$ Mass Concentration

For the contemporaneous $PM_{2.5}$ mass samples at each site (N = 196 in Atlanta, N = 185 in Philadelphia, and N = 120 in Baltimore), $PM_{2.5}$ mass concentrations were compared between each pair of samplers (Figure 5). Linear regressions of the comparison are summarized in Table 2, considering both regular and zero intercept regressions.

In Atlanta, the strongest correlation was observed between the CAMM and SES TEOM (r^2 = 0.78, slope = 0.75 ± 0.03; intercept = 2.51 ± 0.51 μg m^{-3}), while rather mild correlation was observed between the adjusted RAMS and SES TEOM (r^2 = 0.52, slope = 0.85 ± 0.06; intercept = 5.34 ± 1.04 μg m^{-3}) or between the adjusted RAMS and CAMM (r^2 = 0.43, slope = 0.91 ± 0.07; intercept = 5.71 ± 1.20 μg m^{-3}). From the one-way analysis of variance (ANOVA) in 95% confidence intervals, no significance was observed in average $PM_{2.5}$ mass concentration of the CAMM and SES TEOM. However, a statistical significance was observed in the average $PM_{2.5}$ mass concentration of the adjusted RAMS and TEOM or of the adjusted RAMS and CAMM.

In Philadelphia, the strongest $PM_{2.5}$ mass correlation was also found between the CAMM and SES TEOM (r^2 = 0.71, slope = 0.60 ± 0.03; intercept = 2.00 ± 0.42 μg m^{-3}). As high $PM_{2.5}$ mass correlation as in Atlanta was observed between the RAMS and SES TEOM (r^2 = 0.63, slope = 0.71 ± 0.04; intercept = 2.51 ± 0.59 μg m^{-3}) or between the RAMS and CAMM (r^2 = 0.55, slope = 0.93 ± 0.06; intercept = 2.44 ±

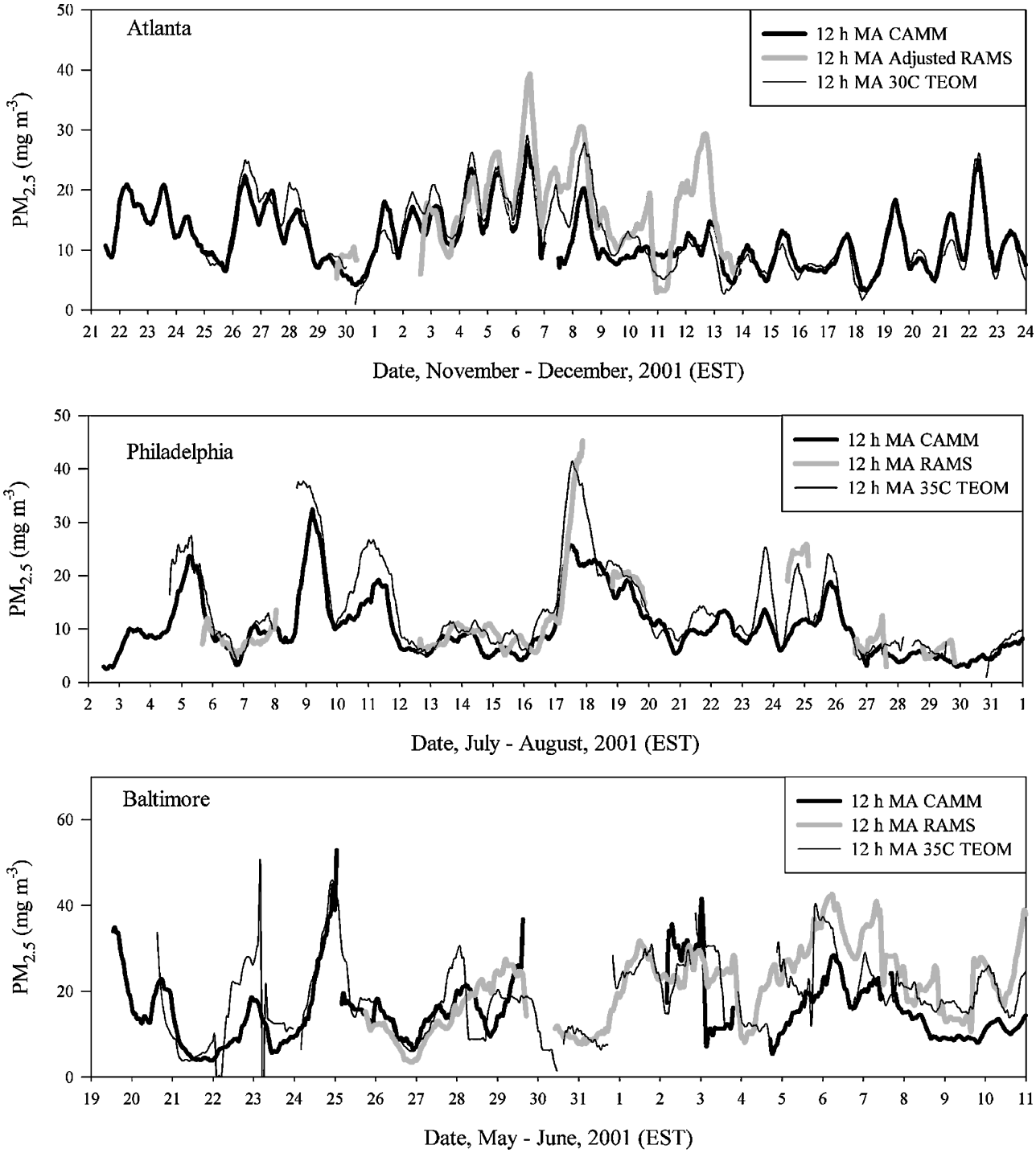


FIG. 4. Twelve hours moving averaged (MA) $PM_{2.5}$ mass concentrations at the sites.

0.68 μg m^{-3}). The average $PM_{2.5}$ mass concentrations of the RAMS and SES TEOM were not statistically different from each other, whereas statistical mass difference was observed both between the CAMM and SES TEOM and between the RAMS and CAMM.

The most unrelevant relationship was found in Baltimore between each pair of the continuous methods showing r^2 values of 0.14–0.38. These low r^2 values in Baltimore imply gross method failures, i.e., not valid method responses to aerosols. The ANOVA results also show that there was a statistically

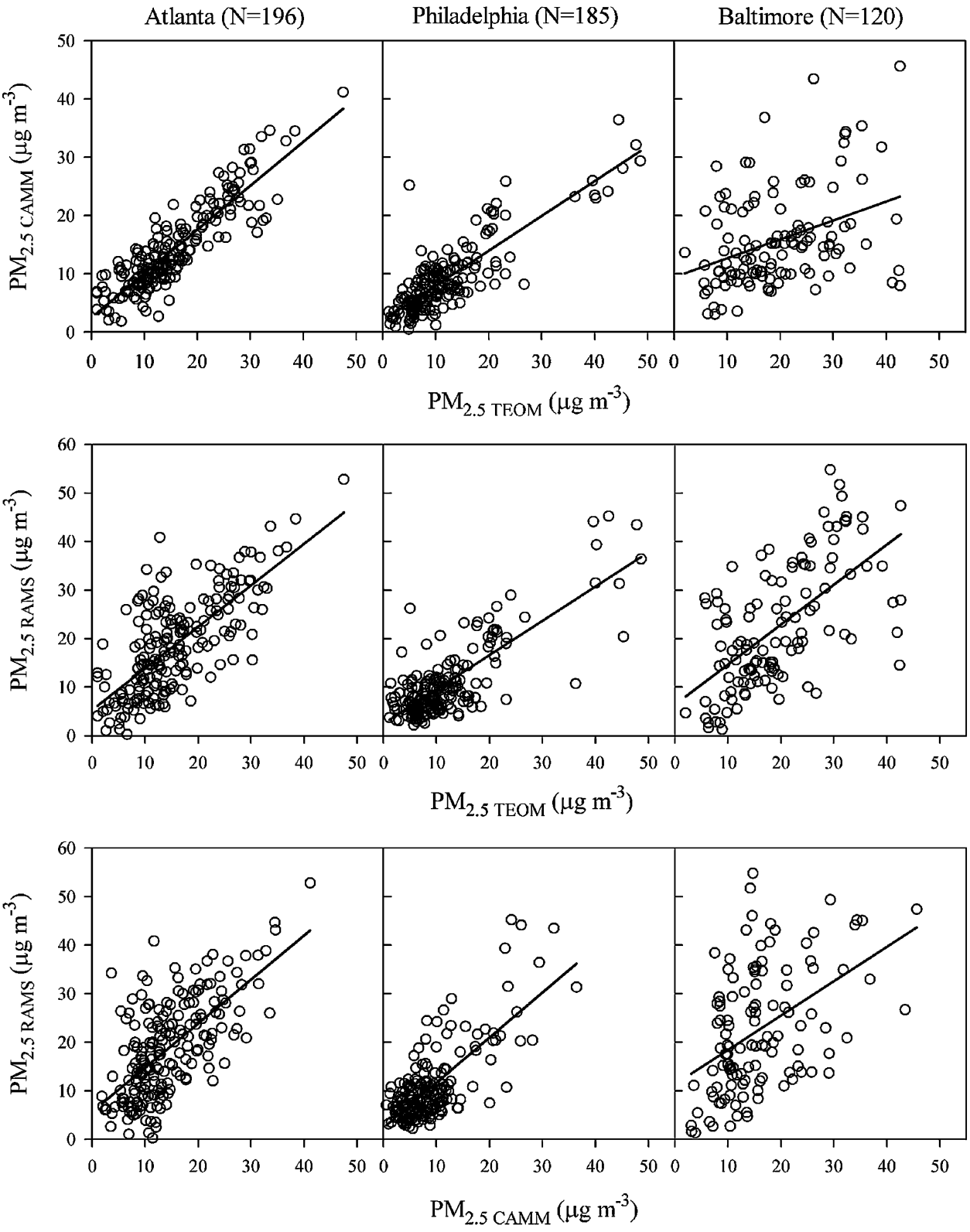


FIG. 5. Intersampler comparison among the CAMM, RAMS, and TEOM $PM_{2.5}$ mass. The RAMS values in Atlanta were adjusted by a factor of 1.64.

TABLE 2
Linear regressions of continuous $PM_{2.5}$ mass concentrations

Method Y	Method X	N	R^2	Slope	Intercept ($\mu g\ m^{-3}$)	Significance ($p < 0.05$)
Atlanta, GA						
CAMM	TEOM	196	0.78	0.75 ± 0.03 (0.88 ± 0.01)	2.51 ± 0.51	No
Adjusted RAMS	TEOM	196	0.52	0.85 ± 0.06 (1.12 ± 0.03)	5.34 ± 1.04	Yes
Adjusted RAMS	CAMM	196	0.43	0.91 ± 0.07 (1.23 ± 0.03)	5.71 ± 1.20	Yes
Philadelphia, PA						
CAMM	TEOM	185	0.71	0.60 ± 0.03 (0.71 ± 0.02)	2.00 ± 0.42	Yes
RAMS	TEOM	185	0.63	0.71 ± 0.04 (0.84 ± 0.02)	2.51 ± 0.59	No
RAMS	CAMM	185	0.55	0.93 ± 0.06 (1.11 ± 0.04)	2.44 ± 0.68	Yes
Baltimore, MD						
CAMM	TEOM	120	0.14	0.32 ± 0.07 (0.71 ± 0.04)	9.45 ± 1.61	Yes
RAMS	TEOM	120	0.38	0.82 ± 0.10 (1.09 ± 0.04)	6.41 ± 2.09	Yes
RAMS	CAMM	120	0.21	0.71 ± 0.12 (1.27 ± 0.06)	11.3 ± 2.23	Yes

Values in parentheses are slopes with the regression lines through origin. The analysis of variance was performed in 95% confidence intervals to examine the statistical difference in average $PM_{2.5}$ mass concentration.

significant difference in the average $PM_{2.5}$ mass between each pair of the methods.

The dissimilarity/similarity in performance between samplers for $PM_{2.5}$ mass concentrations can be expressed by the coefficient of divergence (COD; Wongphatarakul et al. 1998). The COD is defined as

$$\mathrm{COD}_{fh} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_{if} - x_{ih}}{x_{if} + x_{ih}}\right)^2},$$

where x_{if} is the ith concentration measured by the fth monitor, f and h represent two monitoring systems, and n is the number of observations. The approach of the COD to zero or one indicates high similarity or no relevance between two samplers, respectively. A low value of COD indicates that the measurement performance between two samplers is similar (i.e., one sampler can be used as an alternative to the other). Each contemporaneous set of 1 h $PM_{2.5}$ mass concentrations

TABLE 3
Similarity (coefficient of divergence, COD) between each pair of samplers for $PM_{2.5}$ mass measurements

	CAMM vs RAMS	CAMM vs TEOM	RAMS vs TEOM	CAMM vs FRM	RAMS vs FRM	TEOM vs FRM
1 h samples						
Atlanta, GA	0.28	0.20	0.26			
Philadelphia, PA	0.28	0.25	0.26			
Baltimore, MD	0.31	0.29	0.27			
24 h samples						
Atlanta, GA				0.09	0.12	0.06
Philadelphia, PA				0.27	0.26	0.14
Baltimore, MD				0.17	0.15	0.06

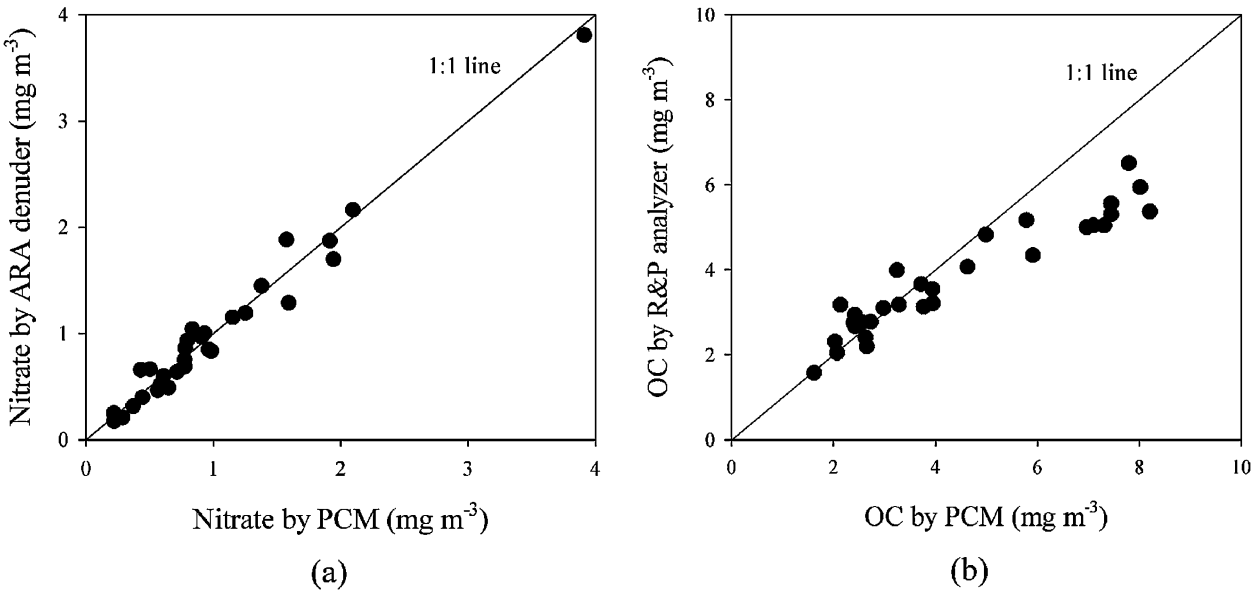


FIG. 6. Continuous versus integrated (a) particulate nitrate and (b) OC concentrations in Atlanta.

were used to calculate the COD values that are shown in Table 3.

The COD values calculated between $PM_{2.5}$ mass samplers were not much different from each set of samplers. In Atlanta, the highest similarity of $PM_{2.5}$ mass measurements was found between the CAMM and SES TEOM with a COD of 0.20. In Philadelphia, the highest similarity of $PM_{2.5}$ mass measurements was also shown between the CAMM and SES TEOM with a COD of 0.25. The COD values between the other sets of samplers (i.e., the CAMM and RAMS or the RAMS and TEOM) ranged from 0.26 to 0.28 in both places. In Baltimore, relatively high COD values were calculated between each pair of $PM_{2.5}$ mass measurements. A highest similarity in $PM_{2.5}$ mass measurements was shown between the RAMS and SES TEOM with a COD of 0.27, and the lowest similarity was shown between the CAMM and RAMS with a COD of 0.31. The range of COD values was smallest in the order of Philadelphia, Baltimore, and Atlanta. These results show that the performance of samplers in PM mass measurement can vary by locations and by seasons because aerosol composition would be expected to be different.

Effects of Semivolatile Losses on $PM_{2.5}$ Mass

Relationship between semivolatile aerosol components and $PM_{2.5}$ mass was further examined using (semi)continuous semivolatile data. Due to insufficient data for semivolatile OC and nitrate, a comparison was made only for those time periods when high nitrate or OC concentrations were seen at each site. Sulfate concentration was not considered in this study because sulfate is not volatile once it is collected on the filter, and the mass difference during and after sampling results mostly from the losses of semivolatile ammonium nitrate and organic material (OM), and particle-bound water. For the Atlanta data, the number of matching hours for $PM_{2.5}$ mass from the three continuous instruments and particulate nitrate was 193, and the average $PM_{2.5}$ mass concentration for those 193 h was 16.2 μg m^{-3}. The comparison was made for those time periods when nitrate concentration was higher than 1.62 μg m^{-3}, that is, 10% of the average $PM_{2.5}$ mass concentration for the 193 h. The time periods with nitrate concentration >1.62 μg m^{-3} were 67 h, for which the average $PM_{2.5}$ mass concentration was 14.3 $\pm$ 0.80 μg m^{-3} for the CAMM, 18.9 $\pm$ 1.10 μg m^{-3} for the adjusted RAMS, and 15.3 $\pm$ 0.93 μg m^{-3} for the SES TEOM. The

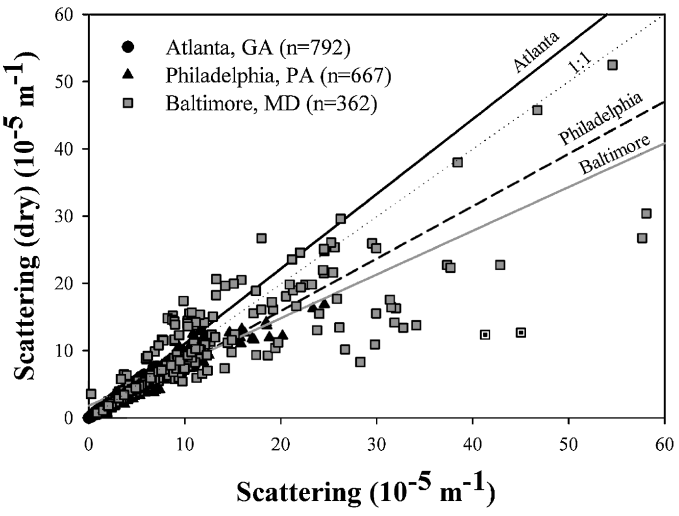


FIG. 7. Scatterplots of particle scattering from nephelometers with and without a drier.

TABLE 4
Comparison of dried and wet light scattering (10^{-5} m^{-1}) in the three eastern US cities

City	r^2	Slope	Intercept
Atlanta, GA	0.996	1.11 ± 0.00	-0.09 ± 0.01
Philadelphia, PA	0.947	0.78 ± 0.01	0.30 ± 0.03
Baltimore, MD	0.748	0.65 ± 0.02	1.80 ± 0.20

average concentration of particulate nitrate and OC for the 67 h period was 2.97 ± 0.15 $\mu g\ m^{-3}$ and 3.45 ± 0.16 $\mu g\ m^{-3}$, respectively. By applying appropriate conversion factors (1.29 for particulate nitrate and 1.6 for OC; Turpin and Lim 2001), concentration of particulate ammonium nitrate and OM is calculated to be 3.83 $\mu g\ m^{-3}$ and 5.52 $\mu g\ m^{-3}$, respectively.

The nitrate analyzers are considered to measure total particulate nitrate, including semivolatile material, while the continuous carbon analyzers (i.e., the R&P carbon analyzer and the Sunset carbon analyzer) are considered not to measure semivolatile material (i.e., nonvolatile OM only). Figure 6 depicts the relationship between the 24 h averaged continuous versus 24 h integrated measurements for particulate nitrate and OC. The integrated nitrate and OC that were measured using the PCM provide total (nonvolatile + semivolatile) particulate nitrate and OC concentration. The 24 h averaged continuous versus integrated nitrate measurements were plotted in Figure 6a, in which a good correlation ($r^2 = 0.97$; slope = 0.97 ± 0.03, intercept = 0.02 ± 0.04 $\mu g\ m^{-3}$) was observed between the two measurements. In the meantime, the 24 h averaged continuous versus integrated OC measurements have a high ($r^2 = 0.90$) correlation, but the slope is smaller than 1 (Figure 6b). The lower slope (0.57 ± 0.04) suggests that SVOM was not measured by the R&P carbon analyzer in Atlanta. The losses of semivolatile particulate ammonium nitrate and SVOM collected on the backup filters of the PCM were 1.12 $\mu g\ m^{-3}$ and 0.57 $\mu g\ m^{-3}$ on average, respectively, which contributes 84% to the total average particulate ammonium nitrate concentration and 7% to the total average OM concentration.

TABLE 5
Results of regression calculated and zero intercept regression (in parentheses)

X versus Y	N	R^2	Slope	Intercept ($\mu g\ m^{-3}$)	X average ($\mu g\ m^{-3}$)	Y average ($\mu g\ m^{-3}$)	X–Y bias ($\mu g\ m^{-3}$)
Atlanta, GA							
FRM versus CAMM	31	0.76	0.75 ± 0.08 (0.93 ± 0.03)	2.47 ± 1.02	12.3	11.8	0.6
FRM versus adjusted RAMS	13	0.64	0.97 ± 0.22 (1.12 ± 0.06)	2.39 ± 3.42	14.8	16.8	−2.0
FRM versus TEOM	26	0.95	1.07 ± 0.05 (0.98 ± 0.02)	-1.34 ± 0.71	13.3	12.8	0.4
Philadelphia, PA							
FRM versus CAMM	22	0.87	0.70 ± 0.06 (0.71 ± 0.02)	0.16 ± 0.96	14.4	10.3	4.1
FRM versus RAMS	3	N/A[a]	N/A	N/A	8.8	7.9	0.9
FRM versus TEOM	15	0.89	1.00 ± 0.10 (0.97 ± 0.04)	-0.68 ± 1.74	16.1	15.5	0.6
Baltimore, MD							
FRM versus CAMM	7	0.60	0.80 ± 0.29 (0.76 ± 0.06)	-0.83 ± 5.85	19.6	14.8	4.8
FRM versus RAMS	11	0.90	1.05 ± 0.12 (1.25 ± 0.05)	4.80 ± 2.60	20.4	26.2	−5.8
FRM versus TEOM	10	0.90	0.86 ± 0.10 (1.00 ± 0.03)	2.96 ± 1.99	19.1	19.5	−0.3

[a]N/A, not available due to insufficient data.

Assuming the same percent contributions to the continuously measured particulate ammonium nitrate and NVOM, semivolatile particulate ammonium nitrate concentration was calculated to be 3.22 μg m^{-3} (84% of total particulate ammonium nitrate concentration, 3.83 μg m^{-3}) and SVOM concentration was calculated to be 0.42 μg m^{-3}. The SVOM was estimated by subtracting NVOM concentration from total OM concentration. Here, total OM concentration (5.94 μg m^{-3}) was approximated by dividing NVOM concentration by a factor of 0.93, based on the ratio of nonvolatile OM to total OM measured by the PCM. As a result, the total SVM concentration becomes 3.64 μg m^{-3} (3.22 μg m^{-3} by semivolatile ammonium nitrate and 0.42 μg m^{-3} by SVOM), which fits approximately well with the mass difference of $PM_{2.5}$ between the adjusted RAMS and SES TEOM for the 67 h. The mass difference of $PM_{2.5}$ between the adjusted RAMS and CAMM was not explained enough by the total SVM concentration. It is probably because the CAMM undermeasured the total ambient $PM_{2.5}$ mass concentration in Atlanta.

In Philadelphia, the number of matching hours for $PM_{2.5}$ mass and OC and nitrate concentration was 36 h. The average $PM_{2.5}$ mass concentration for those 36 h was 14.8 $\pm$ 1.29 μg m^{-3} for the CAMM, 16.1 $\pm$ 1.80 μg m^{-3} for the RAMS, and 20.0 $\pm$ 1.94 μg m^{-3} for the SES TEOM. The average OC and nitrate concentration by the continuous monitors were 9.5 $\pm$ 0.49 μg m^{-3} and 0.5 $\pm$ 0.13 μg m^{-3}, respectively. It should be noted that the OC measured by the Sunset carbon analyzer also does not include semivolatile organic material (i.e., nonvolatile OC only). The nitrate measured by the Dasgupta system observed the total nitrate in a lower level, about 1/3 of the HEADS nitrate concentration. The correlation (r^2) between the two nitrate measurements was 0.72 (not shown). The average sulfate concentration for this time period was 4.7 $\pm$ 0.51 μg m^{-3}. If a reasonable conversion factor (e.g., 1.6; Turpin and Lim 2001) is used to estimate OM, the CAMM data taken during this time period is lower than the sum of OM and ammonium nitrate inferred by the continuous species monitors. The sulfate will not be negligible, and thus the sum of reconstructed compounds would become even higher than the total mass of the RAMS during the time period. The results suggest that both the CAMM and RAMS underrmeasured the total ambient $PM_{2.5}$ mass during this time period.

In Baltimore, the average mass concentration of $PM_{2.5}$ for the 25 h period with high nitrate concentration >1.0 μg m^{-3} was 19.9 $\pm$ 2.11 μg m^{-3} for the CAMM, 22.6 $\pm$ 2.24 μg m^{-3} for the RAMS, and 20.1 $\pm$ 2.09 μg m^{-3} for the SES TEOM. The mass difference between the instruments could result from the semivolatile losses of ammonium nitrate and OM. The further examination of semivolatile components data was not available in this study due to lack of supporting data. It would appear that the total mass of the CAMM is relatively lower than the total mass of the RAMS. The result suggests that the CAMM also underestimated the total ambient PM mass in Baltimore.

Effects of Water on $PM_{2.5}$ Mass

Scattering from the nephelometers may provide an insight about the influence of particle-bound water on $PM_{2.5}$ mass concentration. It should be pointed out that the RAMS and SES TEOM data would not be expected to include particle-bound water because both instruments have the Nafion dryers and heat the filters. The CAMM also uses a diffusion dryer to reduce particle-bound water. Since the CAMM frequently had lower PM mass than both the RAMS and SES TEOM (Figure 4) in Philadelphia and Baltimore, it would not appear that water was retained.

Figure 7 illustrates particle scattering of nephelometers with and without a drier prior to the inflow air stream at three sites. The correlation coefficients, slopes, and intercepts of the regressions between the ambient and dried scattering values are shown in Table 4. Theoretically, dry scattering should be lower than ambient scattering. However, the observed dry scattering was slightly higher than the ambient scattering in Atlanta. It is suspected that the drier did not work well during the study period in Atlanta. In Philadelphia, particle-bound water likely contributed about 20% to fine particle mass concentrations since the slope of the regression line was 0.78. A significant effect (approximately 35%) of particle-bound water was observed with a lower slope of 0.65 and correlation (r^2 = 0.75) in Baltimore where high PM concentrations were noticeable. The high difference of particle-bound water impacts in Baltimore may be partially due to ambient conditions during the study period. Baltimore was cooler during the sampling period (May–June 2001; average temperature = 20°C, average RH = 59%) and the difference between the ambient temperature and the temperature inside the trailer where nephelometers were located was lower. However, Philadelphia during the study period (July– August 2001; (average temperature = 30°C, average RH = 80%) was warmer and more humid. Because an air conditioner was running inside the trailer for other equipment, the temperature inside was much lower then the ambient temperature. Lower temperature controlled lower water vapor, which resulted in a lower RH (53%) for the wet nephelometer than the ambient RH (80%). Therefore, the particle-bound water became lowered in the Philadelphia aerosol samples, whereas the Baltimore aerosol samples contained a relatively significant amount of particle-bound water. It may explain the significant reduction of particle-bound water in Baltimore, where high PM mass concentrations were observed. The results suggest that ambient water contributed significantly to the PM mass and thus is not negligible in the total ambient PM mass measurements.

Comparison with Integrated $PM_{2.5}$

For the comparison with the integrated $PM_{2.5}$ mass data, 1 h PM mass concentrations were converted to 24 h average PM mass concentration at the three locations. The 24 h average $\pm$ standard error $PM_{2.5}$ mass concentrations measured from the FRM were 12.7 $\pm$ 0.8 μg m^{-3} (N = 33) in Atlanta,

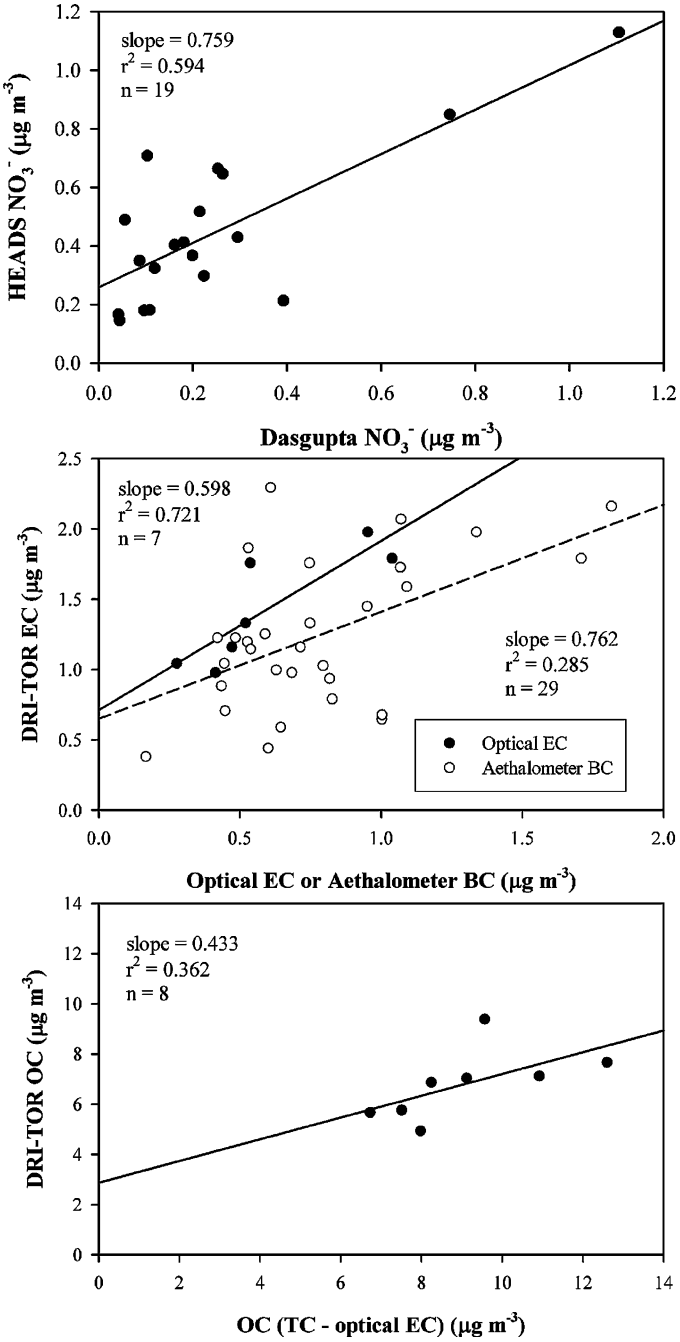


FIG. 8. Comparison of continuous and integrated $PM_{2.5}$ mass concentrations.

17.1 ± 1.7 μg m^{-3} (N = 28) in Philadelphia, and 20.6 ± 2.3 μg m^{-3} (N = 12) in Baltimore.

The similarity in $PM_{2.5}$ mass measurements between the continuous and integrated FRM monitors was examined using the COD. Those COD values in Table 3 were calculated from the matching paired samples of each of the continuous PM mass monitors and integrated mass monitor at each site. The largest similarity between the paired continuous and integrated samplers was seen between the SES TEOM and FRM in all locations. It may be because the SES TEOM can have the same semivolatile losses during sampling as seen in the FRM. In Atlanta, the COD value between the CAMM and FRM was comparable to that between the TEOM and FRM, while the value between the adjusted RAMS and FRM was largest. In Philadelphia, the similar COD values were seen for the pair of the CAMM and FRM and the pair of the RAMS and FRM. The same pattern as for the Philadelphia data set was seen for the Baltimore data set.

Figure 8 depicts the relationship between the $PM_{2.5}$ mass concentrations measured by continuous and integrated samplers in each city. A simple data validation approach was applied by visually examining the scatter plots of the paired PM mass and removing extreme outliers. The regression values were given in Table 5 for both a regression calculated with an intercept and a regression constrained to zero intercept. In Atlanta, the strongest correlation was observed between the SES TEOM and FRM, with a slope of 1.07 and r^2 of 0.95. A stronger correlation was found between the CAMM and FRM ($r^2 = 0.76$, slope = 0.75), but the slope of the regression was higher between the adjusted RAMS and FRM ($r^2 = 0.64$, slope = 0.97). The results show that the total PM mass was underestimated by the CAMM. Similar results were obtained in Philadelphia. The paired SES TEOM and FRM had a higher slope, equal to 1.0, than the paired CAMM and FRM (slope = 0.7), but a similar correlation coefficient ($r^2 = 0.89$) was calculated for both pairs. The CAMM also showed a slope lower than 1, suggesting the CAMM underestimated the total PM at the site. Because of insufficient RAMS data, no comparison was made between the RAMS and FRM. In Baltimore, the best correlation was observed between the RAMS and FRM with a slope of 1.05 and r^2 value of 0.90 that is different from the other two sites. A similar correlation, but a lower slope was observed in Baltimore between the SES TEOM and FRM $PM_{2.5}$ mass concentration. The regression between the CAMM and FRM appeared to be weaker compared to the other pairs of samplers.

The SES TEOM showed the strongest agreement ($r^2 = 0.89$–0.95; slope = 0.86–1.07) with the integrated FRM in ambient $PM_{2.5}$ mass measurements in all three locations. This result seems reasonable since the SES TEOM is also expected to lose semivolatile materials from the filter during sampling as in the FRM. This result is consistent with other studies (Eatough et al. 2003; Lee et al. 2005). The RAMS (adjusted RAMS in Atlanta) mass also showed a reasonable agreement with the FRM mass, but the ambient PM mass was underestimated by the FRM in Atlanta and Baltimore. The bias between averages of the RAMS and FRM mass was higher in Baltimore where summer samples were taken. The RAMS uses two Nafion driers at low dew point and would definitely be expected not to measure water. Thus, the high RAMS data at Baltimore do not include any effect of water. The results are reasonable since the RAMS measures total ambient mass including semivolatile

materials, but the FRM collects nonvolatile PM mass only. The semivolatile losses of ammonium nitrate and organic materials would be higher in warmer period. The sampling season in Atlanta was winter, while the season in Baltimore was summer. Therefore, the volatilization losses of semivolatile materials is expected higher in the Baltimore samples, which explains the higher bias in average mass between the RAMS and FRM in Baltimore. In the same manner, the semivolatile loss would be expected in the TEOM mass measurement. The results of this study are consistent with other studies (Long et al. 2003).

The CAMM showed a consistently lower mass concentration than the FRM, which was significant in Philadelphia and Baltimore. The period that aerosol samples were taken at both sites was summer season. The humid and warm weather prevailed in both locations relative to Atlanta, where samples were taken in cold season. There are two explanations possible for the apparently low mass by the CAMM. One reason is that the CAMM could lose semivolatile ammonium nitrate and organic materials during the time period. Although the CAMM has been designed to measure ambient total mass in real time, the CAMM did not measure semivolatile materials as expected. This failure might be because none of the devices to collect semivolatile materials were involved in the CAMM. Thus, the loss of semivolatile organic and inorganic materials could occur during sampling by the CAMM since short time-sampling intervals do not guarantee the measurement of total ambient PM mass (Lee et al. 2005). Therefore, depending on the season and location, the CAMM might not provide the total ambient mass concentration. The other reason is that there could be an apparent lack of water retention by the CAMM. In the humid period the FRM is expected to retain particle-bound water. However, as much particle-bound water would not be retained by the CAMM as the FRM because of the Nafion dryer.

In summary, the RAMS appeared to measure total ambient PM, including semivolatile materials, while the CAMM and SES TEOM appeared not to be performing as expected for the measurement of semivolatile materials.

CONCLUSION

The ambient $PM_{2.5}$ mass concentrations were measured using the real-time $PM_{2.5}$ mass samplers such as the CAMM, RAMS, and SES TEOM in Atlanta (winter 2001) and in Philadephia and Baltimore (summer 2001). Semivolatile nitrate and OC samples were also taken by the continuous species monitors at each site, along with particle-scattering measurements by nephelometers. Continuous particulate nitrate analyzer appeared to measure total nitrate, including semivolatile nitrate, while the commercial continuous carbon analyzer did not take semivolatile organic material into measurement.

Based on the results of intersampler comparisons of continuous $PM_{2.5}$ mass and components measurements, the loss of semivolatile ammonium nitrate accounted for the mass difference between the adjusted RAMS and the other two monitors in Atlanta. Such an estimation was not taken into consideration in Philadelphia and Baltimore because of the lack of supporting measurement data. However, the mass differences were observed between the RAMS and either of the CAMM or the SES TEOM, and this mass difference could be accounted for by the loss of semivolatile ammonium nitrate and organic material.

The effects of particle-bound water appeared significant, and they contributed approximately 20% to total aerosol mass in Philadelphia. There was higher reduction of particle-bound water in Philadelphia due to the operating conditions at the site. A significant contribution of particle-bound water to the aerosol mass was observed in Baltimore.

A particularly low total PM mass concentration was measured by the CAMM in all locations, either in comparison with the other continuous mass monitors or with the integrated FRM mass monitor. The results suggest that the CAMM undermeasured the ambient total PM mass in this study, probably because it could not efficiently measure the SVMs. The SES TEOM mass showed the best agreement with the integrated FRM mass. The results are consistent with results from other studies that the TEOM loses semivolatile aerosol materials during sampling as in the FRM.

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